

SyncDB Tool



Summary

The SyncDB tool allows administrators to sync the HP AC database with Active Directory. The tool is a command line program that is designed to check every user in the database and delete them (and their aliases) if they match the following conditions in AD:

- Users in the HP AC database which are not listed in AD anymore
- Users in the HP AC database with no email address assigned
- Users in the HP AC database flagged as “Disabled” in AD

IMPORTANT: The customer is responsible for backing up the database before running this tool. The customer is also responsible for restoring the backup in case some database records are unintentionally deleted.

How to use the SyncDB tool

The default syncdb.exe.config file contains the following information:

```
<appSettings>
  <add key="ldapquery" value=
    " (& (& (& (objectCategory=Person) (objectClass=user) (sAMAccountName={0})
    ((mail=*)) (|
    (useraccountcontrol=512) (useraccountcontrol=544) (useraccountcontrol=66048) ) ) ) "
    />
  <!--
  <add key="ldapquery" value=" (& (objectClass=*) (uid={0})) " />
  <add key="ldapserver" value="LDAP://ldap.forumsys.com:389/dc=example,dc=com" />
  <add key="login" value="cn=read-only-admin,dc=example,dc=com" />
  <add key="password" value="password" />
  <add key="simplebind" value="true" />
  <add key="hpacserver" value="localhost" />
  -->
</appSettings>
```

- “ldapquery” is the only tag that is mandatory in the file. The default value will work for AD.
- “ldapserver” is the AD/LDAP server. If the line is commented, the value is read from HP AC. If the line is uncommented, the value is read from the config file. If the ldapserver is specified in the command line, the value is read from the command line.
- “login” is the AD/LDAP login. If the line is commented, the value is read from HP AC. If the line is uncommented, the value is read from the config file. If the login is specified in the command line, the value is read from the command line.
- “password” is the AD/LDAP password. If the line is commented, the value is read from HP AC. If the line is uncommented, the value is read from the config file. If the password is specified in the command line, the value is read from the command line.
- “simplebind” can be set to true false. If the line is commented, the value is read from HP AC. If the line is uncommented, the value is read from the config file. The value can be specified via command line with “true” or “false”. If not specified, “false” will be used.
- “hpacserver” is the name of the HP AC server where the configuration is read. It needs to be changed if the SyncDB tool is not running on the HP AC server. The default value is “localhost”. The value cannot be specified via command line.

NOTE: The tool also does an AD lookup using an LDAP filter that is defined in the SyncDB.exe.config file. The LDAP filter can be changed to query different conditions in AD.

```
<add key="ldapquery" value="
(&(&(&(objectCategory=Person)(objectClass=user)(sAMAccountName={0})((mail=*))(|
(useraccountcontrol=512)(useraccountcontrol=544)(useraccountcontrol=66048)))))" />
```

NOTE: The default LDAP query works only when using Active Directory and not any other version of LDAP, such as OpenLDAP or Tivoli LDAP.

NOTE: Global Catalog works only if there is a guarantee that the user is unique across all domains in the Global Catalog.

The tool is a command line with the following syntax:

```
SyncDB apply [ldapserver] [login] [password] [true|false]
```

All parameters are optional.

NOTE: If the tool is scheduled with the Windows Task Scheduler it must be started with a domain account.

When run with no parameters, the tool will connect to the local Domain Controller and only show what users are active or inactive. No users will be deleted as this is a “test” mode.

When run with the “apply” parameter, the tool shows what users are active or inactive and then delete the users/aliases from the database.

The [Idapserver] parameter can be entered manually if you do not want to do an AD lookup on the full AD. The [Idapserver] parameter must start with “LDAP://” or “GC://”.

For example:

```
SyncDB "LDAP://DC=lrsinc,DC=org"
```

The `[login]` and `[password]` parameters are optional if the user logged into the desktop does not have AD permissions. By default, every domain user will work, but these parameters can be used in case you want to connect to a different domain that is not trusted.

After the command is run, the tool will output results of the sync. This can be redirected to a file using the standard redirect command by adding "`> output.txt`" after the entered parameters.

Below is an example of the output in test mode, run from a desktop connecting to a separate domain, showing valid and invalid users.

```
C:\dev\PullPrintingServices\SyncDB\bin\Release>SyncDB.exe "LDAP://DC=CTQA,DC=LOCAL"
12/13/2018 7:38:52 AM Connecting to HPAC server
12/13/2018 7:38:52 AM Connecting to database
12/13/2018 7:38:52 AM Connected to database
12/13/2018 7:38:52 AM Connected to LDAP://DC=CTQA,DC=LOCAL
12/13/2018 7:38:52 AM 5 Users in database
12/13/2018 7:38:53 AM 1 User BERTRAND is valid
12/13/2018 7:38:53 AM 2 User deJesus is valid
12/13/2018 7:38:53 AM 3 User WOODFIEL is not valid
12/13/2018 7:38:53 AM 4 User CHARBONE is not valid
12/13/2018 7:38:53 AM 5 User BLANCHEW is not valid
```

And below is an example of the output with the “apply” parameter, where users/aliases are deleted.

```
C:\dev\PullPrintingServices\SyncDB\bin\Release>SyncDB.exe apply "LDAP://DC=CTQA,DC=LOCAL"
2/13/2018 7:39:56 AM Connecting to HPAC server
2/13/2018 7:39:56 AM Connecting to database
2/13/2018 7:39:56 AM Connected to database
2/13/2018 7:39:57 AM Connected to LDAP://DC=CTQA,DC=LOCAL
2/13/2018 7:39:57 AM 5 Users in database
2/13/2018 7:39:57 AM 1 User BERTRAND is valid
2/13/2018 7:39:57 AM 2 User deJesus is valid
2/13/2018 7:39:57 AM 3 User WOODFIEL is not valid
2/13/2018 7:39:57 AM 3 User WOODFIEL deleted
2/13/2018 7:39:57 AM 3 User WOODFIEL aliases deleted
2/13/2018 7:39:57 AM 4 User CHARBONE is not valid
2/13/2018 7:39:57 AM 4 User CHARBONE deleted
2/13/2018 7:39:58 AM 4 User CHARBONE aliases deleted
2/13/2018 7:39:58 AM 5 User BLANCHEW is not valid
2/13/2018 7:39:58 AM 5 User BLANCHEW deleted
2/13/2018 7:39:58 AM 5 User BLANCHEW aliases deleted
```

How to use the SyncDB tool with HP AC 15.x

When using HP AC 15.x the tool cannot read information from HP AC. So if the tool is used with HP AC 15.x, it is mandatory to uncomment the lines in the config file, or specify the parameters in the command line.

```
<!--<add key="ldapquery" value="(&(&(&(objectCategory=Person)(objectClass=user) {  
<appSettings>  
  <!--<add key="ldapquery" value="(&(&(&(objectCategory=Person)(objectClass=user) {  
    <add key="ldapquery" value="(&(&(objectClass=*) (uid={0}))" />  
    <add key="simplebind" value="true" />  
    <add key="ldapserver" value="LDAP://ldap.forumsys.com:389/dc=example,dc=com" />  
    <add key="login" value="cn=read-only-admin,dc=example,dc=com" />  
    <add key="password" value="password" />  
    <add key="hpacserver" value="localhost" />  
  </appSettings>
```

How to encrypt/decrypt the configuration file

To encrypt information in the SyncDB.exe.config file, enter the following in a command line:

```
SyncDB encrypt
```

To decrypt the information in order to make changes, enter the following in a command line:

```
SyncDB decrypt
```

NOTE: The file can only be decrypted on the same server it was encrypted.